Robotics · Semester 3 · Group B9

Model Predictive Control for Automated Reverse Docking

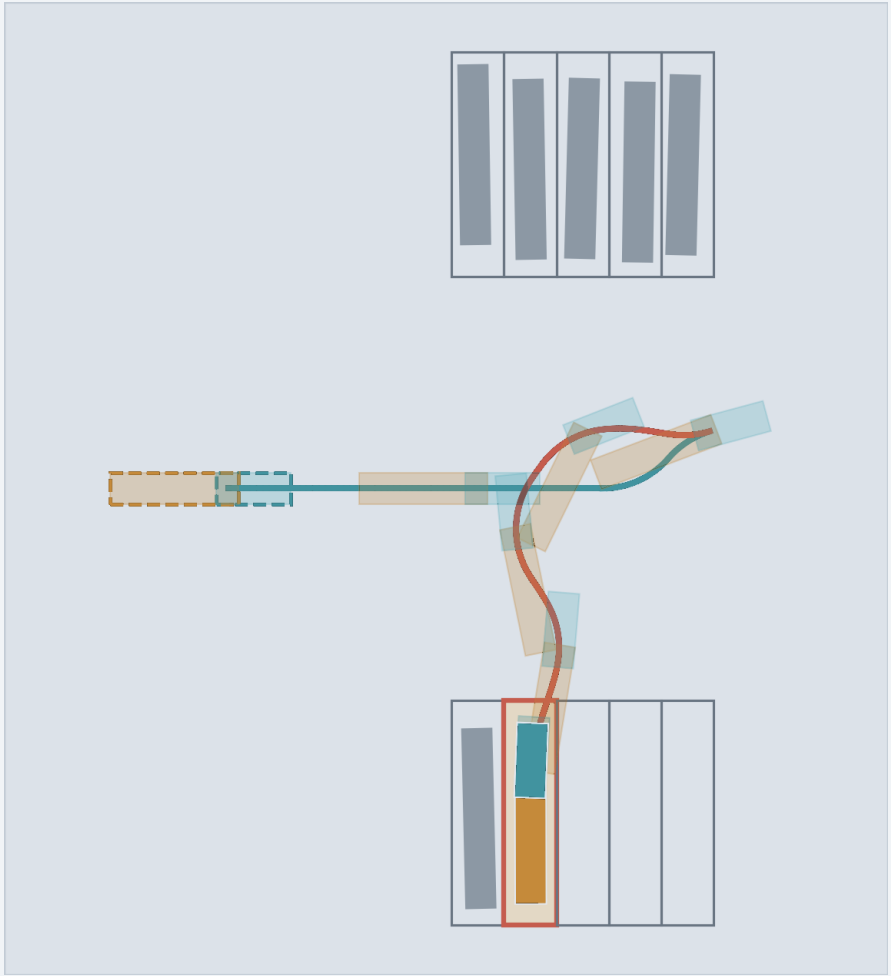
Jackknife-safe parking guidance for a tractor-trailer, using Hybrid A* path planning and nonlinear MPC

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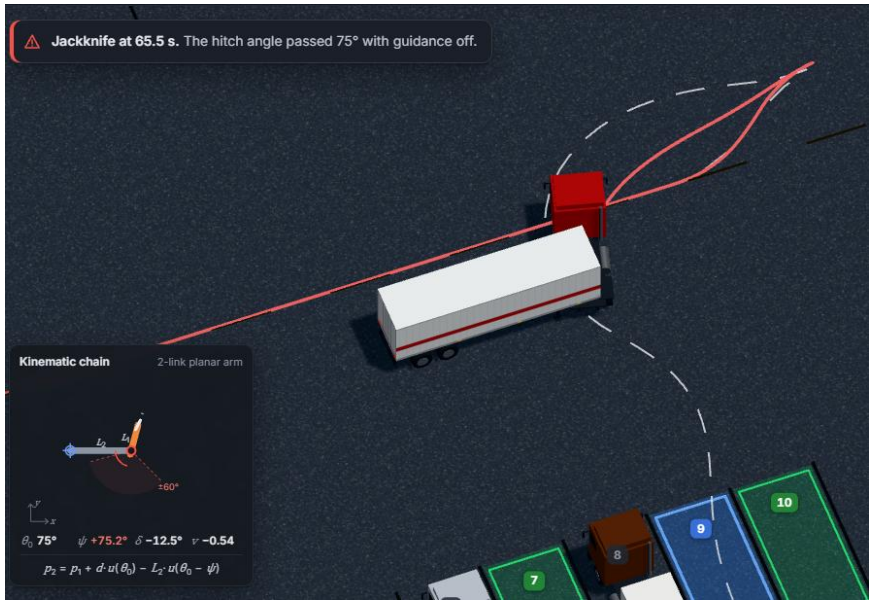
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A real route from our planner: drive in, set up, reverse into the bay

Why a Truck With a Trailer Is a Different Problem

	Car	Tractor-Trailer
Rigid Bodies	1	2, joined at a hitch
State Variables	3	4
Reversing	Stable	Unstable: jackknifes
Self-Parking Today	Yes	No



Unaided reverse in our simulator: the rig folds

Our Goal: guide the truck from the gate into a chosen bay while the hitch angle never exceeds 60°

How This Presentation Is Organised

1

Modelling

The truck as a robot: bodies, joint, states and inputs

2

Vehicle Kinematics

How it moves: the equations and where they come from

3

Hybrid A* and NMPC

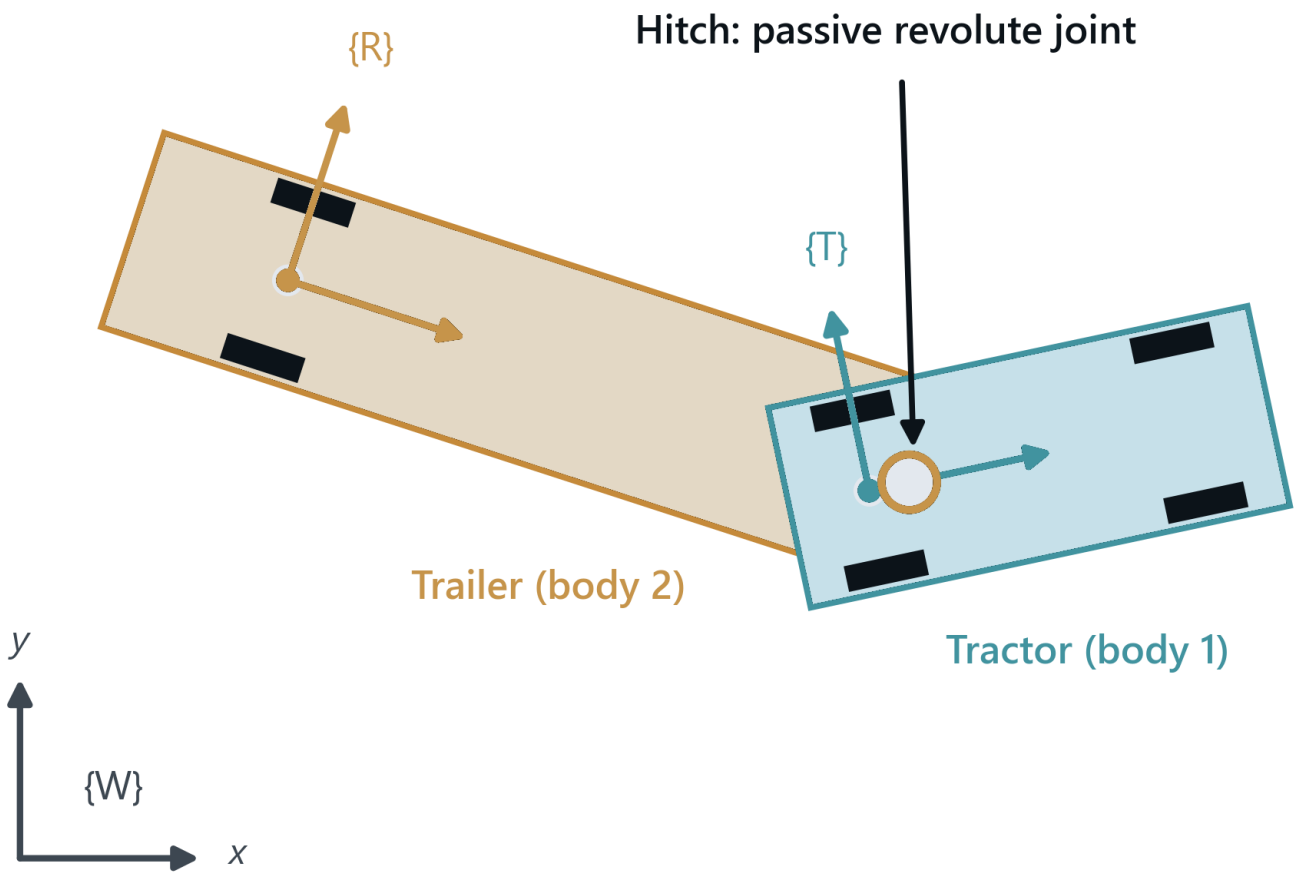
How we plan the route and keep the trailer stable

4

Limitations and Future Scope

Results, what is still missing, what comes next

A Tractor-Trailer Is a Wheeled Mobile Robot



- Two rigid bodies: the tractor and the trailer
- Joined by one passive revolute joint, the hitch
- Its configuration needs four numbers:
  $q = (x_1,\ y_1,\ \theta_0,\ \theta_1)$

Only two inputs are driven: speed and steering. The hitch just follows

Three Robotics Principles Behind the Model

1

Rigid-Body Transforms

Hitch and trailer poses follow from the tractor pose: a chain of frames, like a 2-link arm

2

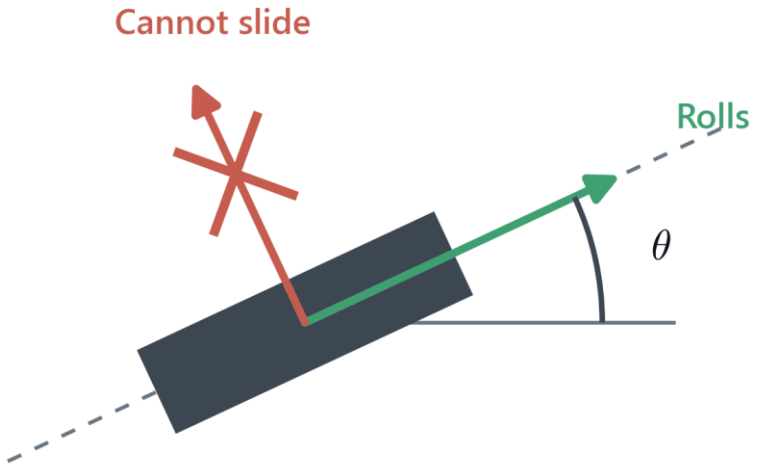
Rolling Without Slipping

Each axle moves only along its own heading, never sideways

3

Nonholonomic but Controllable

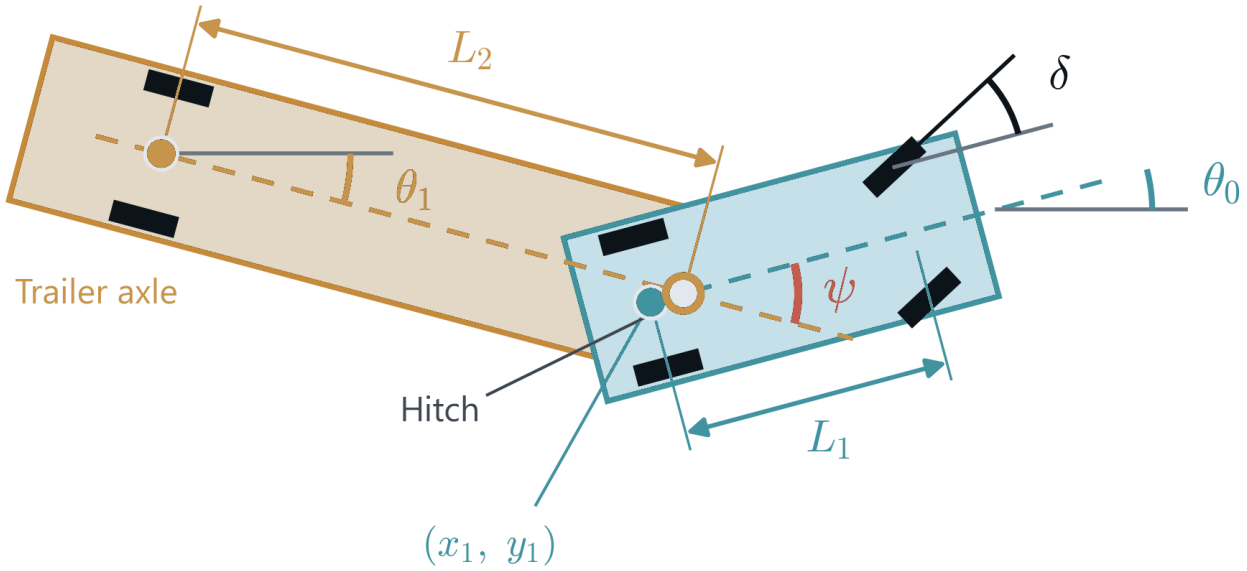
2 inputs still reach all 4 states: that is why parking manoeuvres work



No-Slip Constraint, One per Axle

$$\dot{x}\sin\theta - \dot{y}\cos\theta = 0$$

State Variables and Control Inputs



State

$$X = [\,x_1,\ y_1,\ \theta_0,\ \psi\,]^T$$

x_1, y_1	Rear-Axle Position
θ_0	Tractor Heading
$\psi = \theta_0 - \theta_1$	Hitch angle

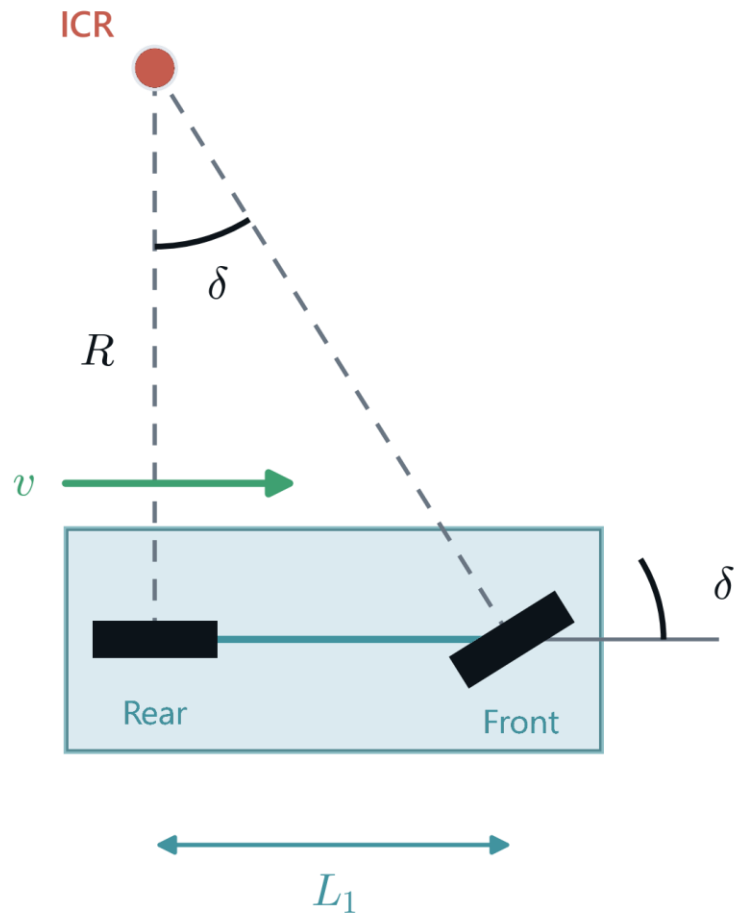
Inputs

$$U = [\,v,\ \delta\,]^T$$

v	Speed (< 0 in reverse)
δ	Steering Angle

$L_1 = 4$ m	$L_2 = 8$ m	$d = 0.5$ m	$\lvert\delta\rvert \le 35°$	$\lvert\psi\rvert \le 60°$	$\lvert v\rvert \le 2$ m/s

The Tractor: Kinematic Bicycle Model



- Front wheels lumped into one, rear wheels into one
- Both wheel axes meet at the instantaneous centre of rotation (ICR)

$$R = \frac{L_1}{\tan\delta}$$

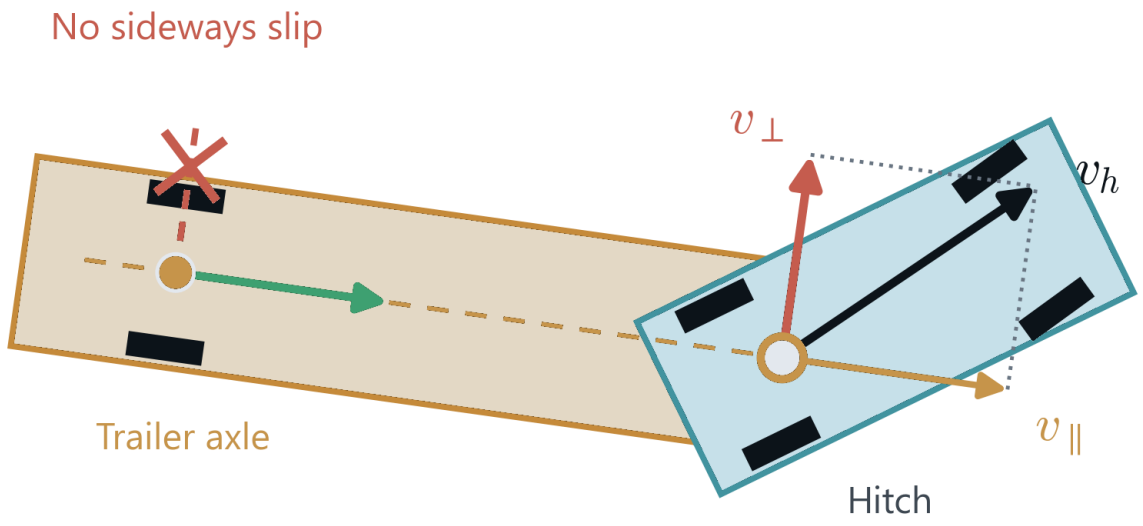
Rear-Axle Motion

$$\dot{x}_1 = v\cos\theta_0$$

$$\dot{y}_1 = v\sin\theta_0$$

$$\dot{\theta}_0 = \frac{v}{R} = \frac{v}{L_1}\tan\delta$$

The Trailer: Hitch Geometry and No-Slip Constraint



1 Hitch Position

$$p_h = p_1 + d\,(\cos\theta_0,\ \sin\theta_0)$$

2 Trailer-Axle Position

$$p_2 = p_h - L_2\,(\cos\theta_1,\ \sin\theta_1)$$

3 Only v⊥ Turns the Trailer

$$\dot{\theta}_1 = \frac{v_\perp}{L_2} = \frac{v\sin\psi + d\,\dot{\theta}_0\cos\psi}{L_2}$$

The Complete Four-State Model

Tractor

$$\dot{x}_1 = v\cos\theta_0$$

$$\dot{y}_1 = v\sin\theta_0$$

$$\dot{\theta}_0 = \frac{v}{R} = \frac{v}{L_1}\tan\delta$$

Hitch Angle

$$\dot{\psi} = \dot{\theta}_0 - \dot{\theta}_1$$

$$\dot{\psi} = v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2} - \frac{d\tan\delta\cos\psi}{L_1 L_2}\right]$$

Steering Term

The only input that acts on the hitch

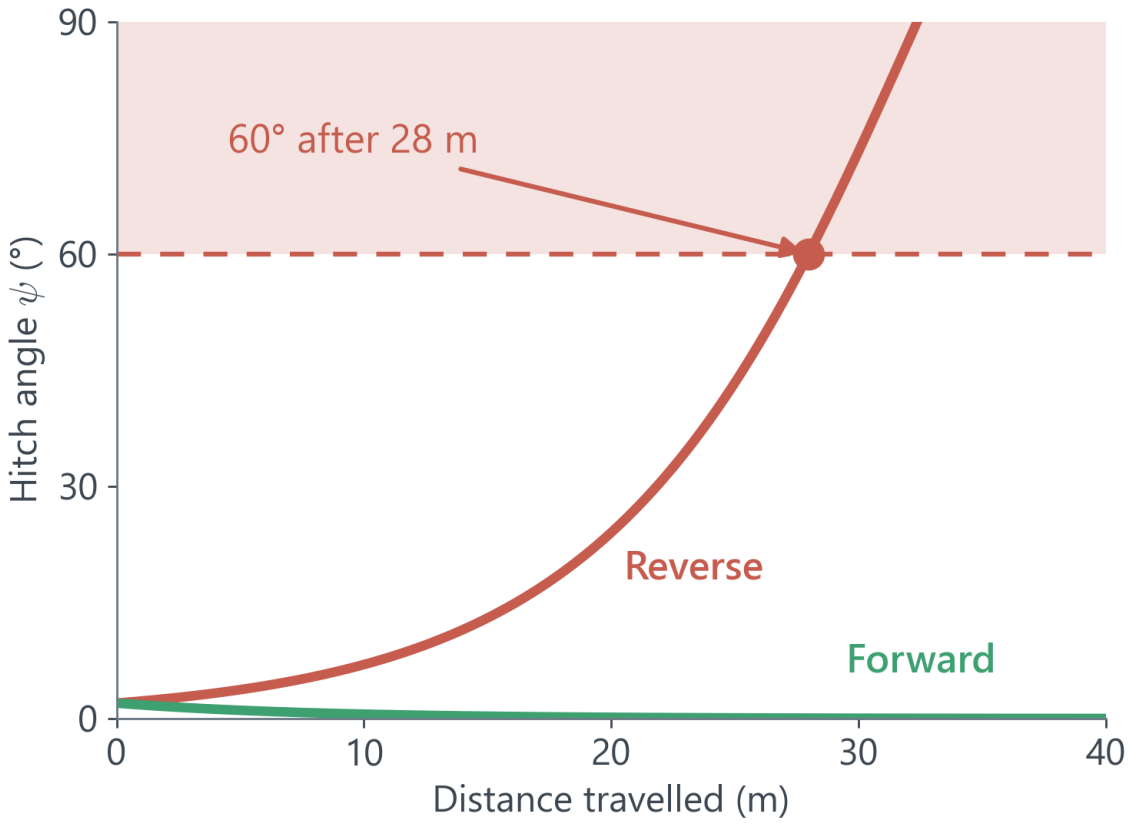
Hitch Term

Restores ψ going forward, destabilises it in reverse

Offset Term

Hitch 0.5 m ahead of the axle: a small correction

Why Reversing Is Unstable



Linearise at $\psi \approx 0$ and $\delta = 0$:

$$\dot{\psi} \approx -\frac{v}{L_2}\psi \quad \Rightarrow \quad \lambda = -\frac{v}{L_2}$$

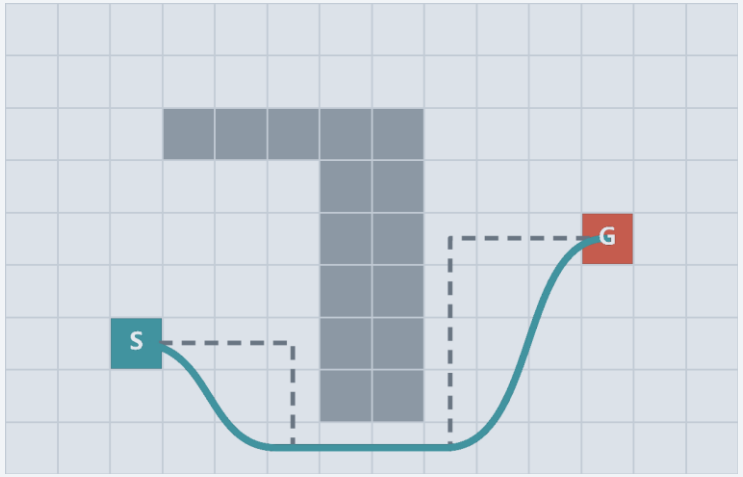
- Forward, $v > 0$: $\lambda < 0$, ψ decays
- Reverse, $v < 0$: $\lambda > 0$, ψ grows
- It grows $\times e$ every $L_2 = 8$ m of travel

28 m

From a 2° error to the 60° limit, in reverse

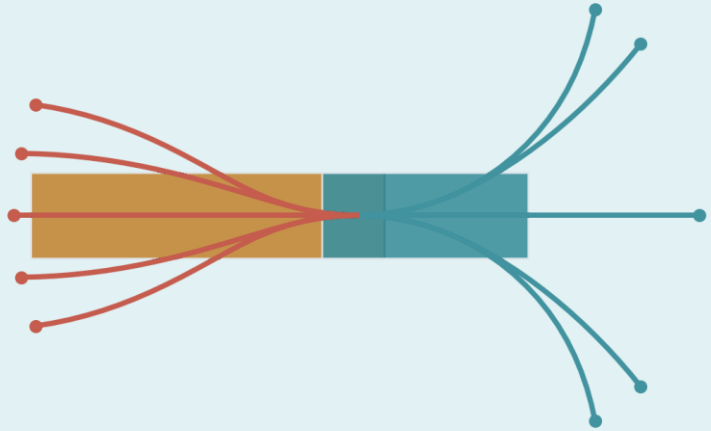
Hybrid A*: Planning a Path the Truck Can Drive

In General



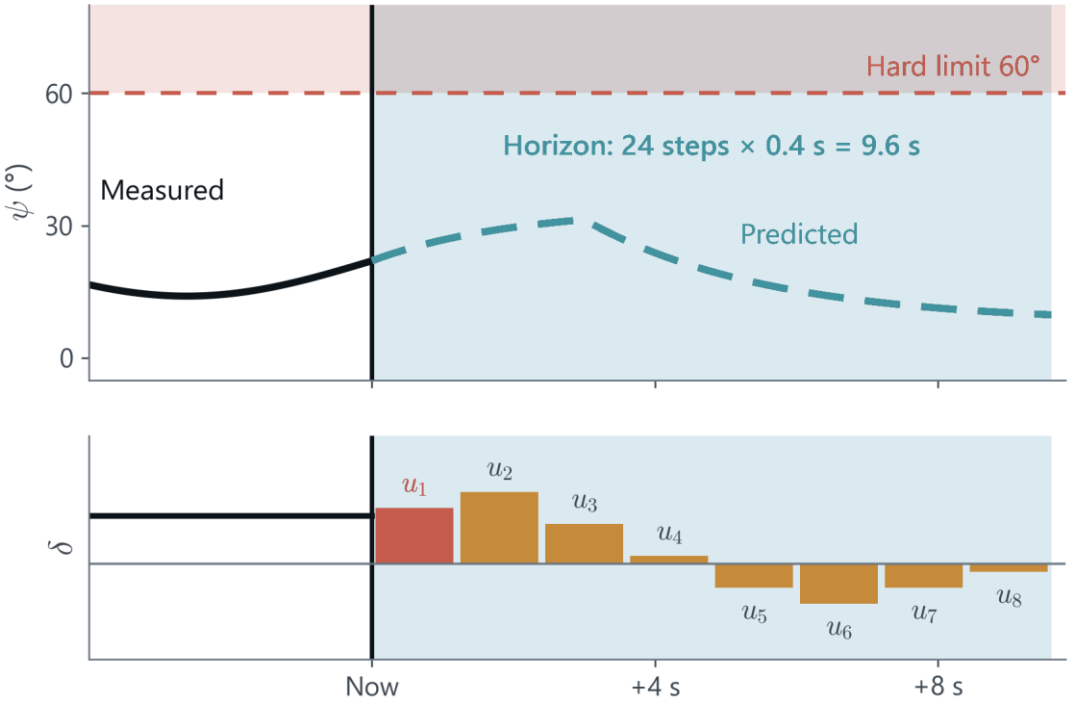
- A* always expands the lowest f = g + h
- Grid paths have corners no truck can drive
- Hybrid A* keeps the exact pose and grows the tree with short drivable arcs

In Our Project



- State (x, y, θ_0, ψ); 10 arcs of 2 m per node, forward and reverse
- Arcs with $|\psi| > 52°$ or under 0.35 m clearance are dropped
- Heuristic: trailer-axle Dubins path [4], [5]

NMPC: Predict, Optimise, Apply, Repeat



In General

1. Predict the next seconds with the model
2. Minimise a cost under hard limits
3. Apply the first input, then repeat

In Our Project

- Horizon 9.6 s, 8 steering moves
- Hard limits on ψ (60°), δ (35°) and clearance
- Re-solved every 0.25 s in 19–42 ms (SLSQP [6])

Cost:

$$J = \sum_{i=1}^{24} \| X_i - X_i^{\mathrm{ref}} \|_Q^2 \; + \; \sum_{j=1}^{8} \left(R\, u_j^2 + S\,(u_j - u_{j-1})^2 \right)$$

How Hybrid A* and NMPC Work Together

Yard and Bay

Obstacles, parked trucks, chosen bay

→

Hybrid A*

Plans the whole route once

→

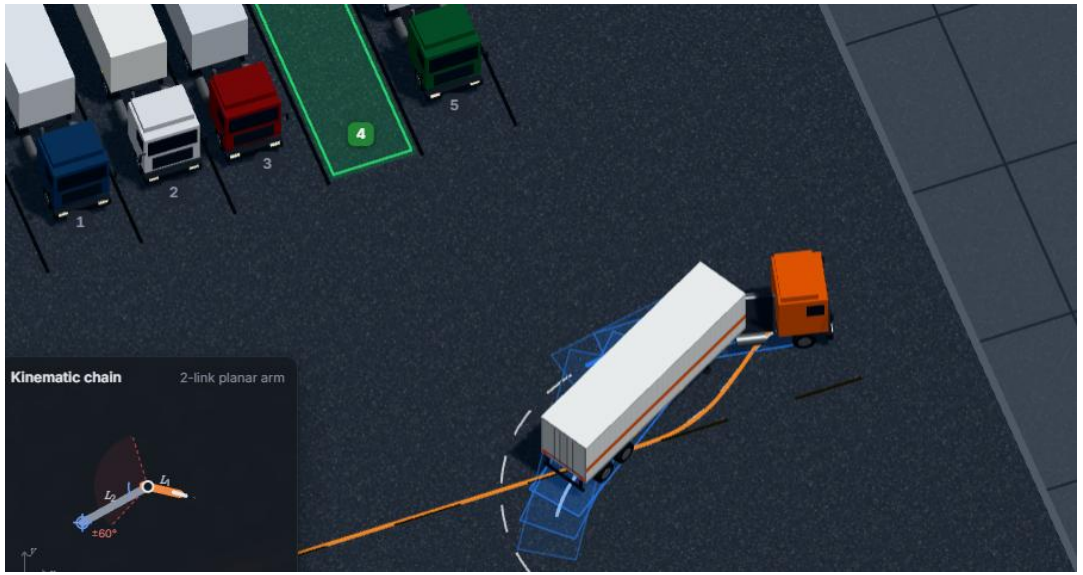
NMPC

Tracks it every 0.25 s, $\psi \leq 60°$

→

Driver Guidance

Gear, km/h, wheel turns, distance



Planned path (dashed) and the NMPC look-ahead as ghost rigs

Planner: Global

Sees the whole yard; decides which way to go and where to switch gear

Controller: Local

Sees 9.6 s ahead; decides how to steer right now and corrects drift

Results at a Glance

110/110

Bays certified for guidance

0

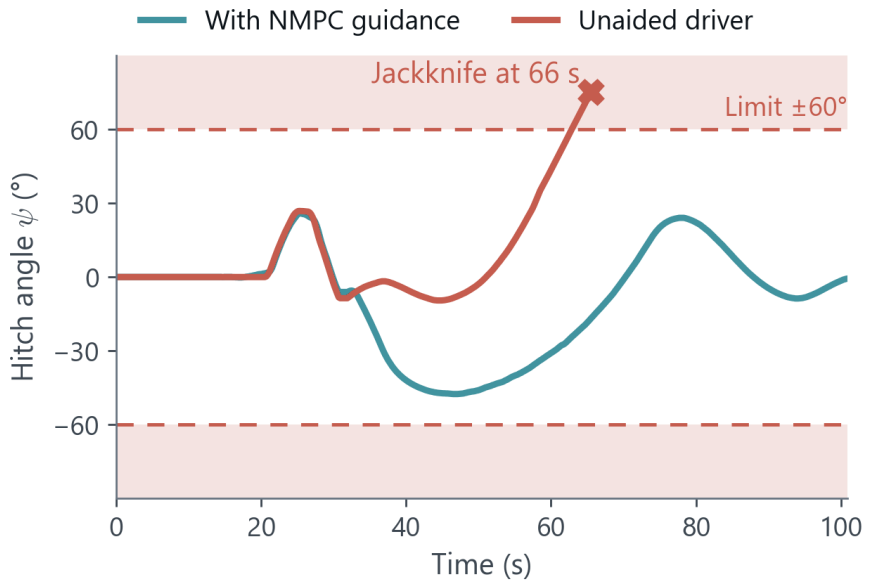
Hitch-limit violations with NMPC

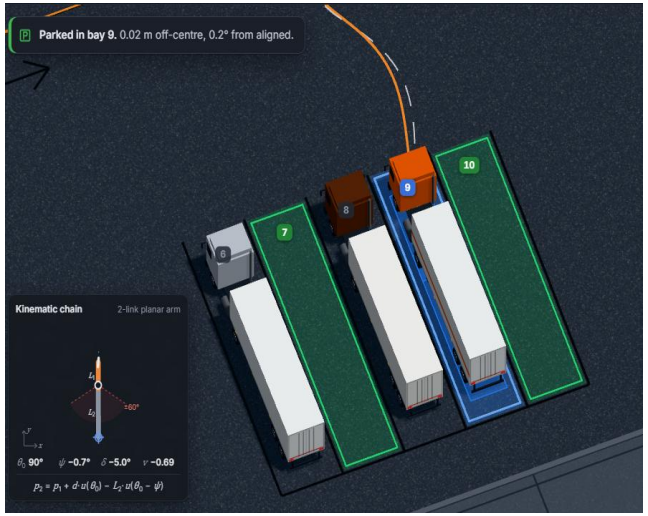
50.1°

Worst hitch angle (limit 60°)

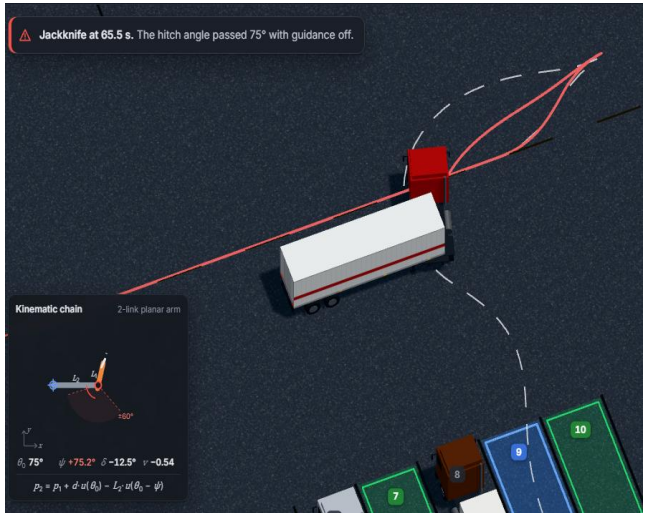
72/72

Unaided reverses jackknife





With NMPC guidance: parked



Without guidance: jackknife

Limitations and Future Scope

Limitations

- Kinematic model: no tyre slip or load transfer
- Pose assumed known: no camera or LiDAR yet
- One trailer, in a simulated yard
- Solver runs on a laptop; tight yards plan slowly

Future Scope

- Camera and LiDAR pose feedback, as base paper [1] proposes
- Faster NMPC (acados, IPOPT) for tighter yards
- Re-planning inside the control loop
- Trials with real drivers; double-trailer rigs

References

[1] G. Alenchery et al., “Parking Assistance for Trailer-Truck Transport Vehicles Using Sensor Fusion and Motion Planning,” arXiv:2605.02716, 2026.

[2] M. Beglini, T. Belvedere, L. Lanari, G. Oriolo, “An Intrinsically Stable MPC Approach for Anti-Jackknifing Control of Tractor-Trailer Vehicles,” IEEE/ASME Trans. Mechatronics, 27(6), 4417–4428, 2022.

[3] E. Kayacan, E. Kayacan, H. Ramon, W. Saeys, “Robust Tube-Based Decentralized Nonlinear Model Predictive Control of an Autonomous Tractor-Trailer System,” IEEE/ASME Trans. Mechatronics, 20(1), 447–456, 2015.

[4] D. Dolgov, S. Thrun, M. Montemerlo, J. Diebel, “Path Planning for Autonomous Vehicles in Unknown Semi-structured Environments,” Int. J. Robotics Research, 29(5), 485–501, 2010.

[5] L. E. Dubins, “On Curves of Minimal Length with a Constraint on Average Curvature,” American J. Mathematics, 79(3), 497–516, 1957.

[6] D. Kraft, “A Software Package for Sequential Quadratic Programming,” DFVLR-FB 88-28, DLR, 1988.

[7] S. M. LaValle, Planning Algorithms, Cambridge University Press, 2006, ch. 13 (nonholonomic systems).